ODISHA ADARSHA VIDYALAYA SANGATHAN (OAVS)

TGT COMPUTER TEACHER EXAMINATION

MCQ Question Bank

Chapter 6: Relational Database

6.1 Introduction to Relational Database

6.2 Codd's 12 Rules (Rule 0 - Rule 12)

6.3 Concept of Domain, Tuple and Cardinality

6.4 Comparison between HDB, NDB and RDB

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For Practice & Preparation Purposes Only

TABLE OF CONTENTS

Introduction to Relational Database 3

Codd's 12 Rules 10

Concept of Domain 25

Concept of Tuple 29

Concept of Cardinality 33

Comparison: HDB vs NDB vs RDB 37

Table-Based & Conceptual Questions 43

Codd's Rule Identification Questions 48

Difficult & Tricky Questions 52

Mixed OAVS TGT Practice Set 56

SECTION 1 — INTRODUCTION TO RELATIONAL DATABASE

21 Questions | Continuous Numbering | OAVS TGT Syllabus 6.1

Q1 **Concept**

The relational database model organizes data in the form of:

A. Tables

B. Hierarchies

C. Graphs

D. Trees

Ans: A

Explanation:
The relational model represents data using two-dimensional tables (relations) consisting of rows and columns.

Q2 **Terminology**

In relational database terminology, a "relation" is most commonly known as a:

A. Field

B. Row

C. Table

D. Domain

Ans: C

Explanation:
A relation is the formal term for what is commonly called a table in a relational database.

Q3 **History**

Who proposed the Relational Database Model?

A. Charles Bachman

B. Peter Chen

C. E. F. Codd

D. Edgar Bell

Ans: C

Explanation:
E. F. Codd, an IBM researcher, proposed the relational model in 1970.

Q4 Terminology

In a relational database, each column of a table represents a(n):

A. Instance

B. Attribute

C. Record

D. Tuple

Ans: B

Explanation:
A column represents an attribute (a characteristic or property) of the relation.

Q5 Terminology

Which term best describes a "row" in a relational table?

A. Attribute

B. Record / Tuple

C. Domain

D. Field

Ans: B

Explanation:
A row of a table is also called a record or tuple, representing one instance of the relation.

Q6 Concept

A key advantage of the relational model over earlier navigational (hierarchical/network) models is:

A. Mandatory pointer chasing for every access

B. Data redundancy is required by design

C. Fixed navigation paths are required

D. Simplicity and flexibility in data access

Ans: D

Explanation:
The relational model allows data to be accessed declaratively without following predefined physical paths, unlike hierarchical/network models.

Q7 Concept

The relational model is mathematically grounded in:

A. Tree traversal algorithms

B. Queueing theory

C. Graph theory

D. Set theory and predicate logic

Ans: D

Explanation:
Codd based the relational model on mathematical set theory and first-order predicate logic.

Q8 Concept

In a relational database, data is fundamentally organized:

A. As a single flat file only

B. In the form of related tables

C. As nested hierarchical trees

D. As linked lists

Ans: B

Explanation:
Data in an RDB is organized into multiple related tables (relations).

Q9 Concept

A limitation noted in early relational systems compared to navigational systems was:

A. Lack of any query capability

B. Inability to store numeric data

C. Inability to represent data in tables

D. Performance overhead for certain operations

Ans: D

Explanation:
Early relational implementations sometimes had performance overhead compared to direct navigational access, though this improved with query optimization.

Q10 Terminology

Which term refers to a "table" in relational database terminology?

A. Relation

B. Field

C. Domain

D. Instance

Ans: A

Explanation:
'Relation' is the formal/technical term for a table in the relational model.

Q11 History

The relational database model was first proposed in the year:

A. 1970

B. 1990

C. 1960

D. 1980

Ans: A

Explanation:
E. F. Codd published his seminal paper introducing the relational model in 1970.

Q12 Terminology

In RDB terminology, a database "schema" refers to:

A. A stored value in a domain

B. A single row of data

C. A single query result

D. The overall logical structure/design of the database

Ans: D

Explanation:
A schema describes the logical structure of the database, including relations, attributes, and constraints.

Q13 Concept

An important characteristic of relational databases is that data is accessed using:

A. Pointer chains only

B. Manual file scanning

C. Declarative, value-based queries

D. Physical storage addresses

Ans: C

Explanation:
Users retrieve data by specifying what they want (declarative queries) rather than how to navigate to it.

Q14 Concept

Which of the following is NOT typically a feature of the relational database model?

A. Data independence

B. Reduced redundancy through normalization

C. Mandatory hierarchical navigation

D. Flexibility in querying

Ans: C

Explanation:
Hierarchical navigation is a feature of hierarchical databases, not the relational model, which uses table-based access.

Q15 Application

A common application area for relational databases is:

A. Managing structured business records such as banking and inventory data

B. Only real-time embedded control systems

C. Only text formatting

D. Only image processing

Ans: A

Explanation:
Relational databases are widely used for structured business data such as banking, inventory, and administrative records.

Q16 Properties

Which of the following is a fundamental property of a relation in the relational model?

A. Each attribute value must be atomic (indivisible)

B. Attributes must be stored in a fixed physical order

C. A relation may contain duplicate tuples

D. Tuples must always be stored in sorted order

Ans: A

Explanation:
A core property of a relation is that every attribute value is atomic (a single, indivisible value), not a list or set.

Q17 Properties

In the relational model, the order of tuples within a relation is:

A. Fixed permanently at the time of table creation

B. Always sorted by the primary key

C. Significant in determining the relation's cardinality

D. Immaterial — it does not affect the relation

Ans: D

Explanation:
Relational theory treats a relation as a set of tuples; since sets are unordered, the sequence in which tuples appear does not change the relation.

Q18 Properties

In the relational model, the order of attributes (columns) within a relation is:

A. Immaterial — it does not affect the relation

B. Must always match the order of data insertion

C. Determines which attribute becomes the primary key

D. Significant for the result of every query

Ans: A

Explanation:
Just as tuple order is immaterial, attribute order within a relation's schema does not change its meaning or identity.

Q19 Properties

According to the fundamental properties of a relation, a relation cannot contain:

A. More than one attribute

B. Two identical (fully duplicate) tuples

C. Numeric attributes

D. Null values in any column

Ans: B

Explanation:
Since a relation is formally a set of tuples, and sets cannot have repeated elements, no two tuples in a relation may be entirely identical.

Q20 Properties

The term "relation schema" refers to:

A. The physical storage location of the table on disk

B. The structure of a relation — its name and the set of attributes it is defined over, independent of actual data

C. A query result derived from the relation

D. A specific set of tuples present at a given moment

Ans: B

Explanation:
A relation schema defines the logical structure (name and attributes) of a relation, as distinct from the data it currently holds.

Q21 Properties

The term "relation instance" refers to:

A. The fixed structural definition of a relation

B. The permissible domain of a single attribute

C. The names of the attributes only, without any data

D. The actual set of tuples present in a relation at a given point in time

Ans: D

Explanation:
A relation instance is the current snapshot of data (the set of tuples) held by a relation, which can change over time even though the schema stays fixed.

SECTION 2 — CODD'S 12 RULES

43 Questions | Continuous Numbering | OAVS TGT Syllabus 6.1

Q22 **Codd's Rule 0**

Codd's Rule 0 (Foundation Rule) states that a system claiming to be relational must:

A. Support hierarchical navigation as the primary access method

B. Manage the entire database exclusively through its relational capabilities

C. Allow bypassing relational operations via low-level commands

D. Store data only in binary format

Ans: B

Explanation:
Rule 0 requires that any system calling itself relational must manage data entirely through relational capabilities.

Q23 **Codd's Rule 0**

Which Codd's Rule is known as the 'Foundation Rule'?

A. Rule 0

B. Rule 5

C. Rule 1

D. Rule 12

Ans: A

Explanation:
Rule 0, the Foundation Rule, is the base requirement upon which all other 12 rules build.

Q24 **Codd's Rule 0**

The Foundation Rule (Rule 0) is significant because:

A. It defines null value handling

B. It establishes the baseline requirement for a system to be called relational at all

C. It defines the physical storage format

D. It relates only to distributed databases

Ans: B

Explanation:
Rule 0 is foundational — without satisfying it, a system cannot be considered truly relational regardless of the other rules.

Q25 Codd's Rule 1

Codd's Rule 1 (Information Rule) states that all information in a relational database must be represented:

A. Implicitly through pointers

B. Explicitly by values in tables

C. Through separate index files only

D. Through hierarchical trees

Ans: B

Explanation:
Rule 1 requires all data — including metadata — to be represented as values within tables, in exactly one way.

Q26 Codd's Rule 1

Which Codd Rule mandates that data be represented only as values in tables (including metadata)?

A. Rule 1

B. Rule 10

C. Rule 4

D. Rule 8

Ans: A

Explanation:
Rule 1, the Information Rule, mandates table-based representation of all data.

Q27 Codd's Rule 1

According to the Information Rule, which of these would violate Rule 1?

A. Storing relationships using separate pointer structures outside tables

B. Storing department codes in a table

C. Storing employee names in a table

D. Storing salary as a numeric column

Ans: A

Explanation:
Rule 1 requires all information, including relationships, to be represented as table values, not via external pointer structures.

Q28 Codd's Rule 2

Codd's Rule 2 (Guaranteed Access Rule) ensures every data item can be accessed using:

A. File name only

B. Row number only

C. Physical disk address only

D. Table name, primary key value, and column name

Ans: D

Explanation:
Rule 2 guarantees logical access to any atomic value via a combination of table name, primary key, and column name.

Q29 Codd's Rule 2

Which Codd rule guarantees that every atomic value is logically accessible?

A. Rule 2

B. Rule 9

C. Rule 6

D. Rule 3

Ans: A

Explanation:
Rule 2, the Guaranteed Access Rule, ensures logical accessibility of each datum.

Q30 Codd's Rule 2

The Guaranteed Access Rule (Rule 2) primarily emphasizes:

A. Encryption of stored data

B. Unambiguous logical addressing of data, not physical location

C. Network-based access to remote servers

D. Physical addressing of data on disk

Ans: B

Explanation:
Rule 2 focuses on logical, not physical, addressing of data.

Q31 Codd's Rule 3

Codd's Rule 3 deals with:

A. Systematic treatment of null values

B. View updating

C. Guaranteed access to data

D. Distribution independence

Ans: A

Explanation:
Rule 3 requires that null values be handled systematically and distinctly from all regular values.

Q32 Codd's Rule 3

According to Rule 3, a null value represents:

A. A numeric zero

B. An empty string

C. Missing or inapplicable information, distinct from zero or blank

D. A default system value

Ans: C

Explanation:
Rule 3 states that nulls represent missing/inapplicable data and must be treated differently from any actual value like zero or an empty string.

Q33 Codd's Rule 3

Which Codd rule number addresses null value handling?

A. Rule 11

B. Rule 7

C. Rule 4

D. Rule 3

Ans: D

Explanation:
Systematic treatment of null values is specified by Codd's Rule 3.

Q34 Codd's Rule 3

A DBMS that treats a null and a blank space as identical values would violate:

A. Rule 2 (Guaranteed Access)

B. Rule 3 (Systematic Treatment of Null Values)

C. Rule 8 (Physical Data Independence)

D. Rule 12 (Non-Subversion)

Ans: B

Explanation:
Rule 3 requires nulls to be distinct from all regular values including blanks or zeros, so equating them violates this rule.

Q35 Codd's Rule 4

Codd's Rule 4 (Dynamic Online Catalog) requires that the database description be:

A. Accessible only to administrators via special tools

B. Stored at the logical level as tables, queryable using the same relational language

C. Hardcoded in application source code

D. Stored only in printed documentation

Ans: B

Explanation:
Rule 4 requires the catalog (metadata) itself to be represented as relational tables, queryable like ordinary data.

Q36 Codd's Rule 4

Which Codd rule is also called the "Dynamic Online Catalog" rule?

A. Rule 10

B. Rule 6

C. Rule 4

D. Rule 1

Ans: C

Explanation:
Rule 4 is known as the Dynamic Online Catalog Based on the Relational Model rule.

Q37 Codd's Rule 4

A system that stores metadata as regular relational tables queryable by users satisfies:

A. Rule 9

B. Rule 2

C. Rule 11

D. Rule 4

Ans: D

Explanation:
This describes Rule 4, which requires metadata to be represented and queried the same way as user data.

Q38 Codd's Rule 5

Codd's Rule 5 (Comprehensive Data Sublanguage Rule) requires the system to support:

A. Only a graphical query builder

B. Multiple unrelated languages for each database task

C. A single language handling data definition, manipulation, integrity constraints, and authorization

D. Only stored procedures

Ans: C

Explanation:
Rule 5 requires at least one comprehensive language (like SQL) covering definition, manipulation, constraints, and transaction control.

Q39 Codd's Rule 5

Which Codd rule is most closely associated with languages like SQL?

A. Rule 5

B. Rule 3

C. Rule 0

D. Rule 8

Ans: A

Explanation:
Rule 5, the Comprehensive Data Sublanguage Rule, is the basis for comprehensive query languages such as SQL.

Q40 Codd's Rule 5

The Comprehensive Data Sublanguage Rule (Rule 5) does NOT require the language to support:

A. Data manipulation

B. Transaction boundaries

C. Data definition

D. Graphical-only, non-textual syntax

Ans: D

Explanation:
Rule 5 specifies a language with well-defined linear syntax; it does not mandate a graphical-only interface.

Q41 Codd's Rule 6

Codd's Rule 6 (View Updating Rule) states that:

A. Only base tables can be queried, not views

B. Views can never be updated

C. All theoretically updatable views must be updatable by the system

D. Views must always be permanently materialized

Ans: C

Explanation:
Rule 6 requires that views which are theoretically updatable must actually be updatable through the system.

Q42 Codd's Rule 6

Which Codd rule concerns the updatability of views?

A. Rule 6

B. Rule 2

C. Rule 7

D. Rule 10

Ans: A

Explanation:
Rule 6, the View Updating Rule, addresses this requirement.

Q43 Codd's Rule 6

A system that allows querying views but never permits updating any view, even simple ones, would risk violating:

A. Rule 5 (Comprehensive Sublanguage)

B. Rule 9 (Logical Data Independence)

C. Rule 1 (Information Rule)

D. Rule 6 (View Updating Rule)

Ans: D

Explanation:
Rule 6 requires updatable views (where theoretically possible) to actually support updates.

Q44 Codd's Rule 7

Codd's Rule 7 requires that insert, update, and delete operations be supported at the:

A. Set level (multiple rows at a time), not just single-row level

B. Single-row level only

C. File level only

D. Byte level only

Ans: A

Explanation:
Rule 7 mandates set-oriented operations, allowing multiple rows to be inserted, updated, or deleted in a single operation.

Q45 Codd's Rule 7

Which Codd rule is known as the "High-Level Insert, Update, and Delete" rule?

A. Rule 11

B. Rule 6

C. Rule 3

D. Rule 7

Ans: D

Explanation:
Rule 7 specifically addresses set-based data manipulation operations.

Q46 Codd's Rule 7

A DBMS that only allows modifying one row at a time (never a set of rows in one command) would violate:

A. Rule 8

B. Rule 12

C. Rule 4

D. Rule 7

Ans: D

Explanation:
Rule 7 requires support for set-at-a-time insert, update, and delete operations.

Q47 Codd's Rule 8

Codd's Rule 8 (Physical Data Independence) ensures that:

A. Applications must be rewritten whenever storage changes

B. Changes to physical storage or access methods do not affect application programs

C. Physical storage can never be changed

D. Changes to table structure never affect stored data

Ans: B

Explanation:
Rule 8 ensures applications remain unaffected when physical storage/access methods change.

Q48 Codd's Rule 8

Which rule addresses independence from changes in physical storage?

A. Rule 10

B. Rule 8

C. Rule 9

D. Rule 5

Ans: B

Explanation:
Rule 8, Physical Data Independence, is concerned with insulating applications from storage-level changes.

Q49 Codd's Rule 8

Adding a new index to speed up queries, without requiring any application code changes, demonstrates:

A. Distribution Independence (Rule 11)

B. Logical Data Independence (Rule 9)

C. Physical Data Independence (Rule 8)

D. Integrity Independence (Rule 10)

Ans: C

Explanation:
Rule 8 covers independence from changes in physical access methods such as indexing.

Q50 Codd's Rule 9

Codd's Rule 9 (Logical Data Independence) ensures that:

A. The logical schema can never be changed

B. Information-preserving changes to base tables do not affect applications

C. Applications must be rewritten if a column is renamed

D. Changes to indexes affect application logic

Ans: B

Explanation:
Rule 9 ensures applications remain unimpaired when information-preserving changes are made to base tables, e.g. splitting or merging tables without data loss.

Q51 Codd's Rule 9

Which Codd rule concerns independence from changes at the logical/table structure level?

A. Rule 4

B. Rule 9

C. Rule 8

D. Rule 2

Ans: B

Explanation:
Rule 9, Logical Data Independence, addresses insulation from structural (logical) schema changes.

Q52 Codd's Rule 9

Splitting one base table into two related tables without affecting existing application queries (via views) demonstrates:

A. Integrity Independence (Rule 10)

B. Non-Subversion (Rule 12)

C. Physical Data Independence (Rule 8)

D. Logical Data Independence (Rule 9)

Ans: D

Explanation:
This is an information-preserving structural change, which Rule 9 protects applications against.

Q53 Codd's Rule 10

Codd's Rule 10 (Integrity Independence) requires that integrity constraints be:

A. Hardcoded separately into every application program

B. Optional and rarely used

C. Definable in the relational sublanguage and stored in the catalog, not embedded in application code

D. Enforced manually by users

Ans: C

Explanation:
Rule 10 requires constraints to be specified declaratively and stored centrally, not scattered across application programs.

Q54 Codd's Rule 10

Which Codd rule ensures integrity constraints are stored in the system catalog rather than application code?

A. Rule 9

B. Rule 6

C. Rule 10

D. Rule 5

Ans: C

Explanation:
Rule 10, Integrity Independence, mandates centralized, catalog-stored constraint definitions.

Q55 Codd's Rule 10

A system where every application must independently re-implement the same "salary > 0" check violates:

A. Rule 11 (Distribution Independence)

B. Rule 7 (High-Level Insert/Update/Delete)

C. Rule 1 (Information Rule)

D. Rule 10 (Integrity Independence)

Ans: D

Explanation:
Rule 10 requires such constraints be defined once centrally in the catalog, not duplicated across applications.

Q56 Codd's Rule 11

Codd's Rule 11 (Distribution Independence) states that the data manipulation sublanguage must remain unchanged:

A. Only for hierarchical systems

B. Only when data is centralized

C. Regardless of whether the data is distributed across multiple locations or centralized

D. Only when using network databases

Ans: C

Explanation:
Rule 11 ensures the same query language works transparently whether data is distributed or not.

Q57 Codd's Rule 11

Which Codd rule addresses transparency of the query language regardless of data distribution?

A. Rule 11

B. Rule 12

C. Rule 4

D. Rule 8

Ans: A

Explanation:
Rule 11, Distribution Independence, is concerned with this transparency.

Q58 Codd's Rule 12

Codd's Rule 12 (Non-Subversion Rule) states that a low-level record-at-a-time interface must not be able to:

A. Access data at all

B. Support batch processing

C. Be used by database administrators

D. Bypass integrity constraints enforced by the higher-level relational language

Ans: D

Explanation:
Rule 12 ensures that any low-level access route cannot be used to circumvent integrity rules defined at the relational level.

Q59 Codd's Rule 12

Which Codd rule prevents low-level access from subverting relational integrity rules?

A. Rule 3

B. Rule 12

C. Rule 7

D. Rule 10

Ans: B

Explanation:
Rule 12, the Non-Subversion Rule, protects the integrity enforcement of the relational level from being bypassed.

Q60 Codd's Rule 12

If a system allows a record-level programming interface to insert data that violates a defined integrity constraint, it violates:

A. Rule 2 (Guaranteed Access)

B. Rule 12 (Non-Subversion Rule)

C. Rule 9 (Logical Data Independence)

D. Rule 6 (View Updating)

Ans: B

Explanation:
Rule 12 exists precisely to prevent this kind of integrity bypass via low-level interfaces.

Q61 Codd's Rules - General

Codd's 12 Rules for relational database systems were published in the year:

A. 1990

B. 1980

C. 1970

D. 1985

Ans: D

Explanation:
While Codd proposed the relational model itself in 1970, he published the 12 Rules separately in 1985 to test how 'truly relational' a DBMS is.

Q62 Codd's Rules - General

The primary purpose of Codd's 12 Rules is to:

A. Define how data should be physically stored on disk

B. Specify the exact SQL syntax every DBMS must use

C. List the hardware requirements for a relational system

D. Define and test the degree to which a DBMS can be considered genuinely relational

Ans: D

Explanation:
Codd formulated these rules as a benchmark to evaluate how closely a database system adheres to the relational model, not as a syntax or hardware specification.

Q63 Codd's Rules - General

According to Codd's classification, a DBMS that satisfies all 13 rules (Rule 0 through Rule 12) is termed:

A. Fully Relational

B. Minimally Relational

C. Partially Relational

D. Pseudo-Relational

Ans: A

Explanation:
A system satisfying every one of Codd's rules is classified as Fully Relational; most commercial systems only partially comply.

Q64 **Codd's Rules - General**

A DBMS that satisfies only the Foundation Rule and Information Rule, but very few of the remaining rules, is often informally described as:

A. Extended Relational

B. Minimally Relational

C. Fully Relational

D. Non-Relational

Ans: B

Explanation:
A system meeting just the basic rules without broader compliance is considered only Minimally Relational under Codd's classification.

SECTION 3 — CONCEPT OF DOMAIN

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Q65 Domain

A "domain" in a relational database refers to:

A. A collection of related tables

B. The set of all permissible/valid values an attribute can take

C. A single specific value stored in a row

D. The name of a table

Ans: B

Explanation:
A domain defines the pool of legal values from which an attribute's values are drawn.

Q66 Domain

The domain of the attribute "Gender" (Male/Female/Other) is an example of:

A. A domain with no permissible values

B. An undefined and unlimited domain

C. A domain with a restricted, defined set of valid values

D. A numeric-only domain

Ans: C

Explanation:
This illustrates a domain restricted to a specific enumerated set of valid values.

Q67 Domain

If the domain of "Age" is defined as integers from 0 to 120, a value of 150 for Age would be:

A. Invalid, as it falls outside the defined domain

B. Valid, since domains have no upper limit

C. Irrelevant to the domain concept

D. Valid, since Age is always numeric

Ans: A

Explanation:
A value outside the specified permissible range violates the domain constraint.

Q68 Domain

The relationship between a domain and an attribute is that:

A. They are always identical concepts

B. A domain draws its values from an attribute

C. A domain has no connection with attributes

D. An attribute draws its values from an associated domain

Ans: D

Explanation:
An attribute is defined over a domain, meaning its values must come from that domain's permissible set.

Q69 Domain

Which of the following is an example of a domain?

A. A single employee's mobile number

B. A row in the Employee table

C. The set of all valid 10-digit mobile phone numbers

D. A table named 'Employee'

Ans: C

Explanation:
A domain is a defined set of valid values, such as all valid 10-digit phone numbers, not a single instance.

Q70 Domain

Two different attributes in the same relation can:

A. Only exist if their domains are different

B. Share the same domain, e.g. 'ShipAddress' and 'BillAddress' both drawing from an Address domain

C. Never share the same domain

D. Only be numeric if they share a domain

Ans: B

Explanation:
Multiple attributes can be defined over the same domain when they represent similar kinds of values.

Q71 Domain

A domain constraint primarily helps ensure:

A. Faster physical storage access

B. Automatic query optimization

C. Unique table names

D. Data validity by restricting attribute values to a permissible set

Ans: D

Explanation:
Domain constraints enforce that only valid, permissible values are stored for an attribute.

Q72 Domain

The domain of the attribute "Marital Status" restricted to {Single, Married, Divorced, Widowed} is an example of:

A. An enumerated (categorical) domain

B. A continuous numeric domain

C. A domain identical to cardinality

D. An unrestricted domain

Ans: A

Explanation:
This is a categorical/enumerated domain with a fixed, finite set of valid values.

Q73 Domain

Which statement about domains is TRUE?

A. A domain always equals the number of rows in a table

B. A domain is the same as a tuple

C. A domain describes possible values; it does not itself represent a specific row of data

D. A domain must always be numeric

Ans: C

Explanation:
A domain is a value pool, distinct from tuples (rows) and from cardinality (row count).

Q74 Domain

For the attribute "Blood Group," which value would violate its typical domain?

A. AB+

B. O-

C. A+

D. Z+

Ans: D

Explanation:
Standard blood group domains include A, B, AB, and O with +/- signs; 'Z+' is not a valid/permissible value.

Q75 Domain

In a relational schema, specifying that "Salary" must be a positive decimal number is an example of defining:

A. The domain of the Salary attribute

B. The cardinality of the Salary attribute

C. The degree of the relation

D. A tuple of the Salary attribute

Ans: A

Explanation:
Restricting Salary to positive decimal values defines its domain (permissible value set).

Q76 Domain

The concept of "domain" in RDB is conceptually similar to which programming concept?

A. A function call

B. A variable name

C. A loop structure

D. A data type with a restricted set of legal values

Ans: D

Explanation:
A domain resembles a data type constraint, defining the legal set of values an attribute may hold.

Q77 Domain

A fundamental characteristic of every domain is that its values are:

A. Always numeric in nature

B. Always identical to the attribute name

C. Always unique across every attribute in the database

D. Atomic — a single value from the domain cannot itself be a list or set of values

Ans: D

Explanation:
Domain values must be atomic (indivisible); a domain cannot hold a composite value such as a list or set as a single entry.

SECTION 4 — CONCEPT OF TUPLE

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Q78 Tuple

In a relational database, a "tuple" refers to:

A. The entire table

B. A single column of a relation

C. A single row of a relation

D. A stored procedure

Ans: C

Explanation:
A tuple is an individual row (record) within a relation.

Q79 Tuple

The number of attributes within a single tuple equals:

A. The cardinality of the relation

B. The number of tables in the database

C. The degree (number of columns) of the relation

D. The domain size

Ans: C

Explanation:
Each tuple contains one value for every attribute (column), so its size matches the relation's degree.

Q80 Tuple

If a "Student" relation has attributes RollNo, Name, and Marks, then a single tuple would represent:

A. The entire Student table

B. The domain of Marks

C. Only the RollNo column across all students

D. One student's complete record with values for RollNo, Name, and Marks

Ans: D

Explanation:
A tuple is a complete row representing one entity instance — here, one student's data across all attributes.

Q81 Tuple

Two tuples in the same relation must:

A. Belong to different relations

B. Never have the same number of attributes

C. Typically be distinct, since a relation is theoretically a set of tuples with no duplicates

D. Always be identical

Ans: C

Explanation:
In strict relational theory, a relation is a set of tuples, and sets do not contain duplicate elements.

Q82 Tuple

Which is an accurate way to identify tuples in a table?

A. Each horizontal row of the table represents one tuple

B. The table name represents a tuple

C. Each vertical column represents one tuple

D. The primary key alone represents a tuple

Ans: A

Explanation:
Tuples correspond to rows (horizontal entries) of a table, not columns.

Q83 Tuple

The relationship between a tuple and a relation is that:

A. A tuple is a set of relations

B. Tuples and relations are unrelated concepts

C. A relation is a set of tuples sharing the same attribute structure

D. A relation contains exactly one tuple

Ans: C

Explanation:
A relation (table) is essentially a collection (set) of tuples, all conforming to the same schema.

Q84 Tuple

If a new record is inserted into a table, this operation adds a new:

A. Tuple

B. Relation

C. Attribute

D. Domain

Ans: A

Explanation:
Inserting a new record adds a new tuple (row) to the relation.

Q85 Tuple

In relational terminology, "tuple" is synonymous with:

A. Table / Relation

B. Column / Field

C. Row / Record

D. Database / Schema

Ans: C

Explanation:
'Tuple' and 'row' (or 'record') are used interchangeably in relational database terminology.

Q86 Tuple

A relation with degree 4 means each tuple contains:

A. 4 tuples

B. 4 attribute values

C. 4 tables

D. 4 domains only

Ans: B

Explanation:
Degree refers to the number of attributes (columns), so each tuple has that many values.

Q87 Tuple

Deleting a tuple from a relation means:

A. Removing one entire row/record from the table

B. Removing a domain constraint

C. Removing one column from the table

D. Removing the whole table

Ans: A

Explanation:
Deleting a tuple removes a single row (record) while the table structure remains.

Q88 Tuple

Which of these best represents a tuple from a "Book" relation (BookID, Title, Author, Price)?

A. BookID, Title, Author, Price (the column headers)

B. (101, 'DBMS Basics', 'A. Sharma', 350)

C. 350 (just the price value)

D. The word 'Book'

Ans: B

Explanation:
A tuple is a complete ordered set of values across all attributes for one record, as shown in the example row.

Q89 Tuple

In set-theoretic terms, a tuple can best be described as:

A. An unordered set of tables

B. A single scalar value

C. A file containing multiple tables

D. An ordered collection of attribute values, one per column of the relation

Ans: D

Explanation:
A tuple is formally an ordered n-tuple of values corresponding to the relation's attributes.

Q90 Tuple

Changing the order in which tuples are physically stored or displayed within a relation:

A. Does not create a new relation — the relation remains logically the same

B. Always results in a logically different relation

C. Always changes the cardinality of the relation

D. Changes the degree of the relation

Ans: A

Explanation:
Since a relation is a set of tuples and sets are unordered, rearranging the storage or display order of tuples does not alter the relation itself.

SECTION 5 — CONCEPT OF CARDINALITY

12 Questions | Continuous Numbering | OAVS TGT Syllabus 6.1

Q91 Cardinality

The "cardinality" of a relation refers to:

A. The number of attributes (columns) in the relation

B. The number of tables in the database

C. The number of tuples (rows) in the relation

D. The number of domains defined

Ans: C

Explanation:
Cardinality specifically counts the tuples (rows) present in a relation.

Q92 Cardinality

If a "Student" table contains 250 rows, its cardinality is:

A. Undefined without knowing the domain

B. The number of columns in the table

C. 1, since it is one table

D. 250

Ans: D

Explanation:
Cardinality equals the row count, so a table with 250 rows has a cardinality of 250.

Q93 Cardinality

When 5 new records are inserted into a relation with an existing cardinality of 40, the new cardinality becomes:

A. 5

B. 45

C. 35

D. 40

Ans: B

Explanation:
Cardinality increases by the number of new tuples added: 40 + 5 = 45.

Q94 Cardinality

Deleting 10 tuples from a relation with cardinality 100 results in a cardinality of:

A. 110

B. 90

C. 100

D. 10

Ans: B

Explanation:
Cardinality decreases by the number of tuples removed: 100 minus 10 = 90.

Q95 Cardinality

Cardinality is distinct from "degree" because:

A. Cardinality counts rows, whereas degree counts columns/attributes

B. They mean exactly the same thing

C. Cardinality applies only to hierarchical databases

D. Cardinality counts columns, whereas degree counts rows

Ans: A

Explanation:
Cardinality (rows) and degree (columns) measure different dimensions of a relation.

Q96 Cardinality

A relation with zero tuples (an empty table) has a cardinality of:

A. Equal to its degree

B. Undefined

C. 0

D. 1

Ans: C

Explanation:
An empty relation still has a defined structure but contains zero tuples, so its cardinality is 0.

Q97 Cardinality

Which scenario correctly illustrates cardinality?

A. A 'Department' table with 8 columns has cardinality 8

B. A 'Department' table with 8 distinct department records has a cardinality of 8

C. A database with 8 tables has cardinality 8

D. An attribute with 8 valid domain values has cardinality 8

Ans: B

Explanation:
Cardinality specifically refers to the row count of a relation, i.e. 8 department records means cardinality 8.

Q98 Cardinality

In everyday exam usage, the cardinality of a relation is best determined by:

A. Counting the number of column headers

B. Counting the permissible domain values

C. Counting the number of rows/records currently present in the table

D. Counting the number of unique tables in the schema

Ans: C

Explanation:
Cardinality is found by counting the tuples/rows in the relation.

Q99 Cardinality

If Table X has cardinality 30 and Table Y has cardinality 20, and no rows are shared, a simple union of their rows (assuming identical structure) would have cardinality:

A. 600

B. 10

C. 20

D. 50

Ans: D

Explanation:
Combining rows from two disjoint tables with the same structure gives a total of 30 + 20 = 50 tuples.

Q100 Cardinality

Which of these does NOT affect the cardinality of a relation?

A. Removing all rows via a bulk delete

B. Inserting a new row

C. Adding a new column (attribute) to the table

D. Deleting an existing row

Ans: C

Explanation:
Adding a column changes the degree, not the cardinality; cardinality changes only with row insert/delete operations.

Q101 Cardinality

A relation's cardinality can change over time mainly because:

A. Tuples are frequently inserted into or deleted from the relation

B. The table is renamed

C. The domain of an attribute changes

D. A new index is created

Ans: A

Explanation:
Since cardinality tracks the current row count, any insert or delete operation changes it.

Q102 Cardinality

A "Customer" relation initially has cardinality 500. After a data cleanup removes 45 duplicate/invalid tuples, the new cardinality is:

A. 545

B. 500

C. 455

D. 45

Ans: C

Explanation:
500 minus 45 = 455 after removing the invalid tuples.

SECTION 6 — COMPARISON: HDB VS NDB VS RDB

18 Questions | Continuous Numbering | OAVS TGT Syllabus 6.1

Q103 HDB vs NDB vs RDB

The Hierarchical Database (HDB) model organizes data in the form of:

A. An interconnected graph/network of records

B. A flat unstructured file

C. A tree structure with parent-child relationships

D. Tables of rows and columns

Ans: C

Explanation:
HDB uses a tree-like structure where each child record has exactly one parent.

Q104 HDB vs NDB vs RDB

In the Hierarchical Database model, a child record can have:

A. Only one parent record

B. Unlimited unrelated parents from different trees

C. Multiple parent records

D. No relationship to any parent

Ans: A

Explanation:
HDB enforces a strict one-to-many parent-child relationship, so each child has exactly one parent.

Q105 HDB vs NDB vs RDB

The Network Database (NDB) model represents data relationships using:

A. Strictly tree-based single-parent hierarchies

B. No relationships between records at all

C. A graph-like structure allowing a record to have multiple owners/parents

D. Two-dimensional tables only

Ans: C

Explanation:
NDB uses a network/graph structure (CODASYL model), allowing many-to-many relationships via multiple owner sets.

Q106 HDB vs NDB vs RDB

A key structural difference between HDB and NDB is that:

A. Neither model supports parent-child relationships

B. Both allow unlimited parents identically

C. NDB allows a record to have multiple parent/owner records, while HDB restricts each child to a single parent

D. HDB allows multiple parents while NDB restricts to one

Ans: C

Explanation:
NDB's network structure supports many-to-many relationships, unlike HDB's strict one-to-many tree structure.

Q107 HDB vs NDB vs RDB

The Relational Database (RDB) model organizes data using:

A. Tree-structured records

B. Two-dimensional tables consisting of rows and columns

C. Sequential flat files only

D. Pointer-based network structures

Ans: B

Explanation:
RDB stores data in tables (relations) made up of rows and columns.

Q108 HDB vs NDB vs RDB

Compared to HDB and NDB, RDB generally offers greater:

A. Flexibility in querying and data access, without needing predefined navigation paths

B. Restriction to single-parent relationships only

C. Rigidity, requiring predefined navigation paths

D. Dependency on physical pointers for every access

Ans: A

Explanation:
RDB allows declarative, flexible queries, unlike HDB/NDB which often rely on predefined navigational paths.

Q109 HDB vs NDB vs RDB

Navigation of data in a Hierarchical Database typically follows:

A. Row-and-column based lookups

B. Random access without structure

C. Any arbitrary declarative query path

D. A fixed top-down path from parent to child nodes

Ans: D

Explanation:
HDB access typically follows the tree's fixed hierarchical path from root to leaf.

Q110 HDB vs NDB vs RDB

Navigation in a Network Database is generally performed via:

A. SQL-style declarative table joins only

B. No navigation is needed at all

C. Following pointers/links between interconnected owner and member records

D. Strict single-parent tree traversal only

Ans: C

Explanation:
NDB relies on navigating pointer-based links among interconnected records (owner-member sets).

Q111 HDB vs NDB vs RDB

Which model is generally considered the most flexible for representing complex, changing relationships?

A. Hierarchical Database (HDB)

B. None of them support complex relationships

C. Network Database (NDB)

D. Relational Database (RDB)

Ans: D

Explanation:
RDB's table-based, declarative approach offers the greatest flexibility for evolving and complex relationships.

Q112 HDB vs NDB vs RDB

A key limitation of the Hierarchical Database model is:

A. Excessive flexibility causing data redundancy

B. Difficulty representing many-to-many relationships directly

C. Requirement of SQL for all operations

D. Inability to represent any relationships

Ans: B

Explanation:
HDB's strict one-parent structure makes representing many-to-many relationships difficult without redundancy.

Q113 HDB vs NDB vs RDB

A key limitation of the Network Database model, compared to RDB, is:

A. Restriction to only one-to-one relationships

B. Increased complexity in navigation and application design due to pointer-based structures

C. Lack of any record structure

D. Complete inability to represent relationships

Ans: B

Explanation:
NDB's reliance on complex pointer navigation makes application development and maintenance more complex than RDB's declarative approach.

Q114 HDB vs NDB vs RDB

Which database model represents data relationships through rows and columns in related tables?

A. Hierarchical Database (HDB)

B. None of the above

C. Relational Database (RDB)

D. Network Database (NDB)

Ans: C

Explanation:
RDB uniquely uses tables (rows and columns) to represent both data and their relationships.

Q115 HDB vs NDB vs RDB

An early example of a Hierarchical Database system is:

A. MongoDB

B. MySQL

C. IBM's IMS (Information Management System)

D. Oracle Database

Ans: C

Explanation:
IBM's IMS is a classic example of a hierarchical database management system.

Q116 HDB vs NDB vs RDB

In terms of data representation, HDB, NDB, and RDB differ mainly in:

A. Whether they can store numeric data

B. How they structure and connect records — as trees, networks, or tables respectively

C. Whether they support any query language at all

D. The type of hardware they require

Ans: B

Explanation:
The fundamental difference among these three models lies in their structural approach to organizing and linking data.

Q117 HDB vs NDB vs RDB

Which of these is an advantage of the Hierarchical Database model?

A. No need for any defined structure at all

B. Fast and simple access when data naturally fits a tree structure

C. Best suited for complex many-to-many relationships

D. Fully flexible ad hoc querying like SQL

Ans: B

Explanation:
HDB performs efficiently for data that naturally follows a strict hierarchical, tree-like structure.

Q118 HDB vs NDB vs RDB

Which of these is an advantage of the Network Database model over the Hierarchical model?

A. Simpler structure than hierarchical databases

B. No pointers are needed

C. Ability to represent many-to-many relationships via multiple owner sets

D. Better suited for purely tree-structured data

Ans: C

Explanation:
NDB can represent many-to-many relationships that HDB cannot handle directly, thanks to its network of owner-member sets.

Q119 HDB vs NDB vs RDB

Among HDB, NDB, and RDB, which model is most suitable for applications requiring flexible, ad hoc queries without predefined access paths?

A. RDB

B. HDB

C. All three are equally suitable

D. NDB

Ans: A

Explanation:
RDB supports declarative, ad hoc querying (e.g. via SQL) without requiring predefined navigation paths, unlike HDB and NDB.

Q120 HDB vs NDB vs RDB

A suitable application for the Hierarchical Database model is:

A. Applications needing flexible many-to-many relationships

B. Applications with no defined data structure

C. General-purpose ad hoc business reporting

D. Systems with naturally tree-structured data, such as organizational charts or file directory systems

Ans: D

Explanation:
HDB suits scenarios where data is naturally hierarchical, like organizational charts or directory trees.

SECTION 7 — TABLE-BASED & CONCEPTUAL QUESTIONS

12 Questions | Continuous Numbering | OAVS TGT Syllabus 6.1

Refer to the following 'Employee' relation for questions in this section:

```
EmpID | Name | Dept | Salary
101 | Ravi | IT | 45000
102 | Meena | HR | 38000
103 | Suresh | IT | 42000
104 | Anita | Finance | 50000
105 | Kiran | HR | 39000
```

Q121 **Table-Based**

In the given Employee relation, the cardinality of the relation is:

A. 5

B. 20

C. 4

D. 1

Ans: A

Explanation:
There are 5 rows/tuples in the table, so cardinality = 5.

Q122 **Table-Based**

In the given Employee relation, the degree (number of attributes) is:

A. 3

B. 4

C. 1

D. 5

Ans: B

Explanation:
There are 4 columns — EmpID, Name, Dept, Salary — so the degree is 4.

Q123 Table-Based

Which of the following represents a single tuple from the Employee relation?

A. EmpID, Name, Dept, Salary

B. 42000 (Salary alone)

C. (103, Suresh, IT, 42000)

D. IT

Ans: C

Explanation:
A tuple is a complete row of values; (103, Suresh, IT, 42000) is one full record.

Q124 Table-Based

The domain of the "Dept" attribute in the Employee relation, based on the given data, includes:

A. {IT, HR, Finance}

B. {45000,38000,42000,50000,39000}

C. {101,102,103,104,105}

D. {Ravi, Meena, Suresh, Anita, Kiran}

Ans: A

Explanation:
The Dept column's observed valid values are IT, HR, and Finance, forming its domain in this instance.

Q125 Table-Based

If a new employee (106, Rohit, Finance, 41000) is inserted into the Employee relation, the new cardinality becomes:

A. 4

B. 6

C. 7

D. 5

Ans: B

Explanation:
Adding one new tuple increases cardinality from 5 to 6.

Q126 Table-Based

Is the value "Marketing" a valid value in the current domain of the "Dept" attribute as shown in the table?

A. Yes, since Dept has no domain restriction

B. No, since only IT, HR, and Finance appear as current domain values

C. Cannot be determined

D. Yes, since any string is valid for Dept

Ans: B

Explanation:
Based on the instance shown, only IT, HR, and Finance are present in the Dept domain; 'Marketing' does not currently appear.

Q127 Table-Based

How many tuples in the Employee relation belong to the "IT" department?

A. 3

B. 4

C. 2 (Ravi and Suresh)

D. 1

Ans: C

Explanation:
Scanning the table, EmpID 101 (Ravi) and 103 (Suresh) both have Dept = IT, giving 2 tuples.

Q128 Table-Based

If EmpID is chosen as the identifying attribute, which value below is consistent with the existing domain pattern for EmpID?

A. 106

B. Ravi

C. 42000.50

D. IT

Ans: A

Explanation:
EmpID values follow a numeric pattern (101–105); 106 is consistent with this domain pattern, unlike text or decimal values.

Q129 Table-Based

After deleting the tuple with EmpID 104 from the Employee relation, the cardinality changes from 5 to:

A. 3

B. 4

C. 6

D. 5

Ans: B

Explanation:
Removing one tuple decreases cardinality by one: 5 minus 1 = 4.

Q130 Table-Based

In the Employee relation, which attribute would most likely have the widest (largest) domain of possible values?

A. Dept, since it is limited to a few categories

B. EmpID, since IDs are sequential and limited

C. Salary, since it can take many possible numeric values

D. Name, since names repeat frequently

Ans: C

Explanation:
Salary can take a very wide range of numeric values, generally giving it the largest practical domain among these attributes.

Q131 Table-Based

Which statement correctly distinguishes cardinality from degree using the Employee relation?

A. Cardinality (4) counts columns; degree (5) counts rows

B. Cardinality and degree are unrelated to this table

C. Cardinality (5) counts rows; degree (4) counts columns — they measure different things

D. Cardinality and degree both equal 5 here

Ans: C

Explanation:
Cardinality refers to the number of rows (5), while degree refers to the number of columns (4).

Q132 **Table-Based**

If two employees in the table happened to have identical values in every column, this would:

A. Violate the relational principle that a relation is a set of distinct tuples

B. Be a normal and expected situation in every relation

C. Have no effect on the relation at all

D. Increase the domain of EmpID

Ans: A

Explanation:
Relational theory treats a relation as a set of tuples, which by definition should not contain exact duplicates.

SECTION 8 — CODD'S RULE IDENTIFICATION QUESTIONS

12 Questions | Continuous Numbering | OAVS TGT Syllabus 6.1

Q133 **Rule Identification**

Given the rule number "Rule 9," identify the rule name.

A. Distribution Independence

B. Physical Data Independence

C. Logical Data Independence

D. Integrity Independence

Ans: C

Explanation:
Rule 9 is officially named the Logical Data Independence rule.

Q134 **Rule Identification**

Given the rule name "Guaranteed Access Rule," identify its rule number.

A. Rule 1

B. Rule 8

C. Rule 5

D. Rule 2

Ans: D

Explanation:
The Guaranteed Access Rule is Codd's Rule 2.

Q135 **Rule Identification**

A rule stating "the database description is represented at the logical level in the same way as ordinary data" describes:

A. Rule 10 – Integrity Independence

B. Rule 6 – View Updating Rule

C. Rule 1 – Information Rule

D. Rule 4 – Dynamic Online Catalog

Ans: D

Explanation:
This description matches Rule 4, which requires metadata to be represented as ordinary relational data.

Q136 Rule Identification

Scenario: A DBMS allows adding a new access path (index) to a table without requiring changes to any existing application program. Which rule does this satisfy?

A. Rule 3 – Null Value Treatment

B. Rule 9 – Logical Data Independence

C. Rule 8 – Physical Data Independence

D. Rule 11 – Distribution Independence

Ans: C

Explanation:
Changes to physical access methods (like indexes) not affecting applications is exactly what Rule 8 requires.

Q137 Rule Identification

Given the rule number "Rule 11," identify its name.

A. Integrity Independence

B. Comprehensive Data Sublanguage

C. Non-Subversion

D. Distribution Independence

Ans: D

Explanation:
Rule 11 is the Distribution Independence rule.

Q138 Rule Identification

A rule requiring "at least one language with well-defined syntax supporting data definition, manipulation, and integrity constraints" describes:

A. Rule 5 – Comprehensive Data Sublanguage Rule

B. Rule 7 – High-Level Insert, Update, Delete

C. Rule 4 – Dynamic Online Catalog

D. Rule 12 – Non-Subversion Rule

Ans: A

Explanation:
This is the defining description of Rule 5, the Comprehensive Data Sublanguage Rule.

Q139 Rule Identification

Given the rule name "Non-Subversion Rule," identify its rule number.

A. Rule 6

B. Rule 3

C. Rule 10

D. Rule 12

Ans: D

Explanation:
The Non-Subversion Rule is numbered Rule 12.

Q140 Rule Identification

Scenario: A programmer uses a low-level API to insert a record that bypasses a defined "not-null" constraint on a column, and the system allows it. This is a violation of:

A. Rule 7 – High-Level Insert, Update, Delete

B. Rule 9 – Logical Data Independence

C. Rule 12 – Non-Subversion Rule

D. Rule 2 – Guaranteed Access Rule

Ans: C

Explanation:
Allowing a bypass of integrity constraints via a low-level interface directly violates Rule 12.

Q141 Rule Identification

Given the rule number "Rule 6," identify the rule name.

A. Integrity Independence

B. Guaranteed Access Rule

C. Foundation Rule

D. View Updating Rule

Ans: D

Explanation:
Rule 6 is the View Updating Rule.

Q142 Rule Identification

A rule stating "null values must be supported to represent missing information, independent of data type, and distinct from all other values" describes:

A. Rule 3 – Systematic Treatment of Null Values

B. Rule 10 – Integrity Independence

C. Rule 1 – Information Rule

D. Rule 2 – Guaranteed Access Rule

Ans: A

Explanation:
This description precisely matches Rule 3.

Q143 Rule Identification

Scenario: The query language used to access a company's centralized database also works unchanged after the database is later distributed across three servers. This satisfies:

A. Rule 11 – Distribution Independence

B. Rule 4 – Dynamic Online Catalog

C. Rule 8 – Physical Data Independence

D. Rule 5 – Comprehensive Data Sublanguage

Ans: A

Explanation:
Rule 11 ensures the data manipulation language remains the same regardless of data distribution.

Q144 Rule Identification

Given the rule name "Integrity Independence," identify its rule number.

A. Rule 9

B. Rule 10

C. Rule 12

D. Rule 7

Ans: B

Explanation:
Integrity Independence corresponds to Codd's Rule 10.

SECTION 9 — DIFFICULT & TRICKY QUESTIONS

12 Questions | Continuous Numbering | OAVS TGT Syllabus 6.1

Q145 **Difficult**

Which of the following statements is FALSE regarding Codd's 12 Rules?

A. No commercially available DBMS fully satisfies all the rules in strict practice

B. The rules were proposed to test the 'relational-ness' of a DBMS

C. All 13 rules (0–12) are equally easy to implement in every commercial DBMS

D. Rule 0 is a prerequisite foundation for the other rules

Ans: C

Explanation:
In practice, few if any commercial DBMS products satisfy every single rule perfectly; the rules vary greatly in implementation difficulty, making this statement false.

Q146 **Difficult**

A student claims: "Domain and Cardinality mean the same thing." This statement is:

A. Correct, since both relate to counting

B. Incorrect — domain refers to permissible values of an attribute, while cardinality refers to the number of tuples in a relation

C. Correct, since both apply only to numeric attributes

D. Partially correct only for primary key attributes

Ans: B

Explanation:
Domain and cardinality are distinct concepts — one concerns valid attribute values, the other concerns row count.

Q147 **Difficult**

Which pair of Codd's rules are most closely related in purpose, both dealing with data independence?

A. Rule 5 and Rule 11

B. Rule 2 and Rule 6

C. Rule 8 and Rule 9

D. Rule 3 and Rule 7

Ans: C

Explanation:
Rule 8 (Physical Data Independence) and Rule 9 (Logical Data Independence) both address insulating applications from underlying database changes, differing only in physical vs. logical scope.

Q148 Difficult

A relation currently has cardinality 0 but degree 5. This means:

A. The table has 5 rows and no columns

B. The table cannot exist in a relational database

C. The domain of every attribute is also 0

D. The table structure with 5 attributes exists, but currently contains no rows/tuples

Ans: D

Explanation:
Degree (structure/columns) and cardinality (row count) are independent; a table can have a defined structure with zero current rows.

Q149 Difficult

Which of the following is a common misconception about Hierarchical Databases (HDB)?

A. That HDB enforces a one-parent-per-child rule

B. That HDB can represent many-to-many relationships as easily as RDB

C. That HDB uses a tree structure

D. That HDB was widely used before the relational model became popular

Ans: B

Explanation:
A common misconception is that HDB handles many-to-many relationships well; in reality, its strict tree structure makes this difficult, unlike RDB.

Q150 Difficult

Rule 6 (View Updating Rule) is often considered one of the hardest Codd rules to fully implement because:

A. It conflicts directly with Rule 1

B. Views are never used in real databases

C. It requires physical storage changes for every view

D. Not all views are theoretically updatable, and determining updatability for complex views is difficult

Ans: D

Explanation:
Complex views (e.g. involving joins or aggregations) often have ambiguous update semantics, making full compliance with Rule 6 technically challenging.

Q151 Difficult

In a relation, if the degree increases from 4 to 5 (a new column added) while cardinality remains 100, this means:

A. The domain of all attributes doubled

B. 100 new tuples were added

C. The relation now has 500 tuples

D. Each of the 100 tuples now has 5 attribute values instead of 4, but the number of tuples is unchanged

Ans: D

Explanation:
Adding a column changes degree, not cardinality; each existing tuple simply gains an additional attribute value.

Q152 Difficult

Which statement correctly differentiates NDB from RDB regarding relationship representation?

A. Both represent relationships identically using SQL joins

B. NDB represents relationships via physical pointers between records; RDB represents relationships via matching values across tables, such as foreign keys

C. RDB requires physical pointers, unlike NDB

D. NDB has no concept of relationships

Ans: B

Explanation:
NDB relies on pointer-based navigation, whereas RDB uses value-based matching (like foreign key references) to represent relationships.

Q153 Difficult

A tricky aspect of Rule 3 (null values) is that:

A. Nulls cannot exist anywhere in a relation without restriction

B. Nulls only apply to numeric attributes

C. Nulls are the same as the value zero

D. A null must be treated distinctly from zero, blank, or any default value, for all data types

Ans: D

Explanation:
Rule 3 requires consistent null handling across all data types, distinguishing null from any valid value including zero or blank — a nuance often misunderstood.

Q154 Difficult

Which of the following would NOT change the cardinality of a relation?

A. Inserting a new tuple

B. Renaming an existing column

C. Bulk-loading 50 new records

D. Deleting an existing tuple

Ans: B

Explanation:
Renaming a column affects the schema/attribute name, not the number of tuples, so cardinality is unaffected.

Q155 Difficult

Assertion (A): Codd's Rule 12 (Non-Subversion) implies that low-level interfaces should be entirely eliminated from a relational DBMS. Reason (R): Low-level interfaces always violate integrity constraints. Which is correct?

A. Both A and R are false — Rule 12 only requires that low-level access not bypass integrity rules; it does not eliminate such interfaces, nor does using one always violate constraints

B. Both A and R are true, and R correctly explains A

C. A is true but R is false

D. A is false but R is true

Ans: A

Explanation:
Rule 12 permits low-level interfaces to exist, but they must not be usable to bypass integrity constraints enforced at the relational level; neither assertion is accurate as stated.

Q156 Difficult

Which scenario best illustrates the difference between HDB, NDB, and RDB using the same data (an employee working in multiple projects)?

A. HDB would struggle to represent an employee linked to multiple projects without data duplication; NDB could link them via network pointers; RDB would typically use a separate linking/junction table

B. HDB naturally supports many-to-many links without any redundancy

C. All three models handle this scenario identically

D. Only RDB can store employee data at all

Ans: A

Explanation:
This scenario highlights the structural differences: HDB's tree structure struggles with many-to-many links, NDB uses pointer-based owner-member sets, and RDB typically resolves this via a junction/linking table.

SECTION 10 — MIXED OAVS TGT PRACTICE SET

13 Questions | Continuous Numbering | OAVS TGT Syllabus 6.1

Q157 **Mixed Practice**

Which of the following is NOT part of Codd's 12 Rules?

A. Non-Subversion Rule

B. Guaranteed Access Rule

C. Normalization Rule

D. View Updating Rule

Ans: C

Explanation:
Normalization is a separate database design concept; it is not one of Codd's 12 rules for relational systems.

Q158 **Mixed Practice**

The total number of Codd's rules, including the Foundation Rule (Rule 0), is:

A. 13 (Rule 0 through Rule 12)

B. 15

C. 12

D. 10

Ans: A

Explanation:
Codd defined 12 numbered rules plus an additional Rule 0 (Foundation Rule), totaling 13 rules.

Q159 **Mixed Practice**

In an "Order" relation with cardinality 320, the term "cardinality" specifically tells us:

A. That there are 320 columns in the table

B. That there are 320 different tables

C. That the domain of OrderID has 320 permissible values

D. That there are 320 order records (tuples) in the table

Ans: D

Explanation:
Cardinality of 320 means the Order relation currently contains 320 tuples.

Q160 Mixed Practice

Which database model would be most appropriate for representing a simple organizational reporting structure where each employee has exactly one manager?

A. Hierarchical Database (HDB)

B. Relational Database only, with no alternative

C. None of these models can represent it

D. Network Database (NDB) only

Ans: A

Explanation:
A strict one-manager, tree-like reporting structure fits naturally into the Hierarchical Database model, though RDB can also represent it via foreign keys.

Q161 Mixed Practice

According to Rule 7, which operation demonstrates set-level processing?

A. Creating a new table

B. Updating exactly one employee's salary by writing a separate command for each employee

C. Reading a single row using its primary key

D. A single UPDATE command that changes the salary of every employee in the IT department at once

Ans: D

Explanation:
A single command affecting multiple matching rows at once exemplifies the set-level operations required by Rule 7.

Q162 Mixed Practice

The attribute "PIN Code" in an Address relation would typically have a domain consisting of:

A. Valid numeric postal codes following the required format/range

B. Any arbitrary text string

C. A single fixed value for all records

D. Only alphabetic characters

Ans: A

Explanation:
PIN Code's domain is restricted to valid postal code values following a specific numeric format.

Q163 Mixed Practice

If Relation A has cardinality 12 and degree 3, how many individual data values (cells) does it contain in total?

A. 36

B. 9

C. 15

D. 12

Ans: A

Explanation:
Total values = cardinality times degree = 12 x 3 = 36.

Q164 Mixed Practice

Which of these correctly matches a Codd's Rule with its core idea?

A. Rule 1 – Information Rule: null values must be handled systematically

B. Rule 1 – Information Rule: all data is represented as values in tables

C. Rule 1 – Information Rule: distribution transparency

D. Rule 1 – Information Rule: views must be updatable

Ans: B

Explanation:
Rule 1 specifically requires all information to be represented explicitly as values within tables.

Q165 Mixed Practice

A "tuple" and a "record" in relational database terminology are:

A. Completely unrelated concepts

B. Essentially synonymous, both referring to a single row of a relation

C. Tuple refers to columns, record refers to rows

D. Tuple applies only to hierarchical databases

Ans: B

Explanation:
In relational terminology, 'tuple' and 'record' are used interchangeably to mean a single row.

Q166 Mixed Practice

Which of the following best distinguishes RDB from NDB in terms of data access?

A. RDB uses declarative, value-based queries; NDB relies on navigational, pointer-based access

B. Both use identical declarative SQL-based access

C. There is no meaningful difference in access methods

D. NDB uses tables; RDB uses pointers

Ans: A

Explanation:
RDB's declarative querying (specifying 'what' is needed) contrasts with NDB's navigational, pointer-following access (specifying 'how' to get there).

Q167 Mixed Practice

A "Course" relation has the domain of "Credits" defined as {1, 2, 3, 4}. Which value would violate this domain?

A. 4

B. 6

C. 2

D. 3

Ans: B

Explanation:
Since the domain only permits values 1 through 4, a value of 6 falls outside the permissible set and is invalid.

Q168 Mixed Practice

According to Codd's Rule 10, where should integrity constraints ideally be defined?

A. In the relational data sublanguage, stored centrally in the system catalog

B. Only in the user manual/documentation

C. Separately within each individual application program

D. Nowhere — constraints are optional and unnecessary

Ans: A

Explanation:
Rule 10 (Integrity Independence) requires constraints to be defined declaratively and stored centrally, not embedded in application code.

Q169 **Mixed Practice**

Which of the following is TRUE about the relationship between degree and cardinality of a relation?

A. Degree is the same as the number of tuples

B. Increasing cardinality always increases degree

C. Degree always equals cardinality

D. Degree and cardinality are independent properties — one can change without necessarily affecting the other

Ans: D

Explanation:
Degree (attribute count) and cardinality (tuple count) are independent measures of a relation.